

Technology #123

Incorporate micro-vibration dampening
materials and structures within pacemaker
lead insulation to protect internal
components from mechanical stress during
patient movement

Engineering Technology-Transfer Dossier

CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION

Generated: 2026-08-30

Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Incorporate micro-vibration dampening materials and structures within pacemaker lead insulation to protect internal components from mechanical stress during patient movement

WHY DOES IT MATTER?

Documented recurring problem 'therapeutic_or_diagnostic_output_failure' for pacemaker lead: 10 MAUDE reports across 3 distinct devices (20230811 to 20230812); root-cause evidence from 1 recall records and 10 event narratives (custody attached in the campaign report). MAUDE limitations: incidence UNKNOWN, causation UNVERIFIED (Art. XXI.5). Existing solutions fail because: Why existing solutions fail, per retrieved evidence: 1 recall root-cause descriptions (FDA enforcement records, custody attached); 10 MAUDE event narratives from this mechanism cluster (mdr_text, custody attached). Statements are the providers' own words, cust Unresolved gap: Across the 7 relevant retrieved attempts (a bounded, non-exhaustive set), none measurably addresses: conditions, devices, events, function, intervention, introducing, maintain, malfunction, matching, modes, must, problem, producing, recurring, reported, under,

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 3 governing equations, 12 design inputs, 6 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in mechanical_structural. 10-step build plan with specific test articles and acceptance criteria. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHAT DOES THE BUYER GET?

Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

WHAT DOES THE BUYER HAVE TO BUILD?

Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

WHAT IS THE NEXT DECISIVE EXPERIMENT?

first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

WHAT WOULD MAKE US KILL IT?

killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 1 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 10-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed) - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: physical bench/clinical experiment — NOT simulation. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - article for VF-001 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: Subject pacemaker leads with and without vibration-damping systems to accelerated mechanical stress testing while monitoring electrical o.... Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - article for VF-003 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: buckling / kink margin tests across duty cycle. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 3 equations. Key assumptions: stress concentrators multiply local stress above the nominal value — geometry factors must be applied from the actual design; superelastic members (e.g. nitinol) do not follow linear elasticity through transformation — use their measured plateau relations instead; K for catheter constraint conditions is MODEL_DERIVED; MEC-002 (Section bending stress) is applicable only under: linear-elastic material; small deflection; prismatic section.; MEC-003 (Hooke's law (linear elasticity)) is applicable only under: below proportional limit; isothermal.; MEC-001 (Euler buckling load) is applicable only under: elastic, slender column, small deflection.; Model input variables requiring sourced values before any numeric result: M (N.m), c (m), I (m⁴), E (Pa), epsilon (-), E (Pa), I (m⁴), K (-), L (m) — all UNKNOWN (no sourced value exists; Art. VIII).; Model output variables produced by the governing model: sigma (Pa), sigma (Pa), P_cr (N) — UNKNOWN until the inputs are sourced and the model is evaluated.; Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): member slenderness ratio [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; yield/ultimate strength [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; applied cyclic load amplitude [P_cr, N] value_status=UNKNOWN; required run-out cycles [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; articulation radius [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; friction coefficient at guides [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

1 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: Vibration-damping systems in automotive suspensions that isolate components from external stressors while maintaining functionality

POTENTIAL VALUE DRIVER: Addresses: Documented recurring problem 'therapeutic_or_diagnostic_output_failure' for pacemaker lead: 10 MAUDE reports across 3 distinct devices (20230811 to 20230812); root-cause evidence from 1 recall records and 10 event narratives (custody attached in the campaign report). MAUDE limitations: incidence UNKNOWN, causation UNVERIFIED (Art. XXI.5). Existing solutions fail because: Why existing solutions fail, per retrieved evidence: 1 recall root-cause descriptions (FDA enforcement records, custody attached); 10 MAUDE event narratives from this mechanism cluster (mdr_text, custody attached). Statements are the providers' own words, cust Unresolved gap: Across the 7 relevant retrieved attempts (a bounded, non-exhaustive set), none measurably addresses: conditions, devices, events, function, intervention, introducing, maintain, malfunction, matching, modes, must, problem, producing, recurring, reported, under,

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: domain-appropriate technology buyer (to be identified by portfolio owner)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: mechanical_structural

Disciplines: mechanical engineering, structural analysis, fatigue and fracture, materials science

Architecture: PROPOSED architecture (no physical system exists): load path / structural backbone; actuation or compliance element; guidance / bearing interfaces; fatigue-critical features register — realizing: Incorporate micro-vibration dampening materials and structures within pacemaker lead insulation to protect internal components from mechanical stress during patient movement

Subsystems:

?: load path / structural backbone - ?

?: actuation or compliance element - ?

?: guidance / bearing interfaces - ?

?: fatigue-critical features register - ?

2. Mechanism Architecture

Physical Changes: PROPOSED: Incorporate micro-vibration dampening materials and structures within pacemaker lead insulation to protect internal components from mechanical stress during patient movement

Key Physics: Mechanics / structures governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

3. Governing Engineering Model

Summary: Mechanics / structures governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Equations:

MEC-002: Section bending stress. Expression: $\sigma = M * c / I$. Applicability to this invention: APPLICABLE; holds under: linear-elastic material; small deflection; prismatic section. Assumptions: stress concentrators multiply local stress above the nominal value — geometry factors must be applied from the actual design. Engaged variables: σ .

MEC-003: Hooke's law (linear elasticity). Expression: $\sigma = E * \epsilon$. Applicability to this invention: APPLICABLE; holds under: below proportional limit; isothermal. Assumptions: superelastic members (e.g. nitinol) do not follow linear elasticity through transformation — use their measured plateau relations instead. Engaged variables: σ .

MEC-001: Euler buckling load. Expression: $P_{cr} = \pi^2 * E * I / (K * L)^2$. Applicability to this invention: CONDITIONAL; holds under: elastic, slender column, small deflection. Assumptions: K for catheter constraint conditions is MODEL_DERIVED. Engaged variables: NOT ESTABLISHED.

Assumptions:

- stress concentrators multiply local stress above the nominal value — geometry factors must be applied from the actual design
- superelastic members (e.g. nitinol) do not follow linear elasticity through transformation — use their measured plateau relations instead
- K for catheter constraint conditions is MODEL_DERIVED
- MEC-002 (Section bending stress) is applicable only under: linear-elastic material; small deflection; prismatic section.
- MEC-003 (Hooke's law (linear elasticity)) is applicable only under: below proportional limit; isothermal.
- MEC-001 (Euler buckling load) is applicable only under: elastic, slender column, small deflection.
- Model input variables requiring sourced values before any numeric result: M (N.m), c (m), I (m^4), E (Pa), ϵ (-), σ (Pa), I (m^4), K (-), L (m) — all UNKNOWN (no sourced value exists; Art. VIII).
- Model output variables produced by the governing model: σ (Pa), σ (Pa), P_{cr} (N) — UNKNOWN until the inputs are sourced and the model is evaluated.

- Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): member slenderness ratio [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; yield/ultimate strength [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; applied cyclic load amplitude [P_cr, N] value_status=UNKNOWN; required run-out cycles [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; articulation radius [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; friction coefficient at guides [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.

Boundary Conditions:

- to be established by the first bench characterization (not yet performed)

Failure Regimes:

- {'regime': 'therapeutic or diagnostic output failure', 'basis': 'SOURCE_FACT (problem.json)'}
- {'regime': 'kink/buckling below required push ability', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'fatigue crack initiation at stress concentrator', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'wear debris generation at sliding interfaces', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'joint separation under overload', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'set-release of superelastic shape after set', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}

4. Design Inputs

ID	Input	Value	Origin
DI-001	Stated constraint	Devices matching 'pacemaker lead' must maintain function under the conditions of occurring malfunction events	SOURCE_FACT
DI-002	Mechanism driver (proposed)	Vibration-damping systems in automotive suspensions that isolate components from external stressors while maintaining	MODELED
DI-003	Expected effect target	Prevent micro-fractures and connection failures in pacemaker leads that lead to interface or diagnostic output issues	MODELED
DI-004	Domain design input pattern: delivery path applicability	NOT_ESTABLISHED (candidate pattern; confirm against evidence)	ENGINEERING_PROPOSED
DI-005	Domain design input pattern: actuation applicability	NOT_ESTABLISHED (candidate pattern; confirm against evidence)	ENGINEERING_PROPOSED
DI-006	Domain design input pattern: duty cycle applicability	NOT_ESTABLISHED (candidate pattern; confirm against evidence)	ENGINEERING_PROPOSED
DI-007	Regulatory design input: compliance candidate standards	ASTM F2516 (ratinal design) pathway UNKNOWN (never inferred)	ENGINEERING_PROPOSED
DI-008	Custodied observation: Bradycardia https://pubmed.ncbi.nlm.nih.gov/31847484/	3rd digest_hash fcf4c98cc1c4f503a67b4935b7c03fd8b198a99fb31edd0	SOURCE_FACT
DI-009	Custodied observation: Takotsubo syndrome https://retire.setpoint.org/article/41857469/	event_digest_hash 3b3a001e8005020f80556a212f765c3e980e7239913d30	SOURCE_FACT
DI-010	Custodied observation: When the Pacemaker Goes Rogue: Practical Aspects of Cardiac Arrest by Chandan Srinivasan et al. 2015	42206028c70d50c080c06879ac5525454f01819b2b4d23fc63	SOURCE_FACT
DI-011	Custodied observation: Sinus Passer and Co. have a dysfunctioning article 42290218	Chetrapatiashipical000aec3807402e06a75d90ee5c0cea4d5448a94bc	SOURCE_FACT
DI-012	Custodied observation: Leadless pacemakers free from precursors article 42955336	guide_to_the_publisher_000aeb8f504a9e77fa30295746d2897a196a1fa803	SOURCE_FACT

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	load path / structural backbone realizing Incorporate micro-vibration dampening	CONCEPTUAL	materials and structures definition
DO-002	actuation or compliance element realizing Incorporate micro-vibration dampening	CONCEPTUAL	materials and structures definition
DO-003	guidance / bearing interfaces realizing Incorporate micro-vibration dampening	CONCEPTUAL	materials and structures definition
DO-004	fatigue-critical features register realizing Incorporate micro-vibration dampening	CONCEPTUAL	materials and structures definition
DO-005	load path / structural backbone interfaces with actuation or compliance element: signal energy amplification (dimensions, tolerances)	CONCEPTUAL	micro-vibration dampening
DO-006	actuation or compliance element interfaces with guidance / bearing interfaces: signal energy amplification (dimensions, tolerances)	CONCEPTUAL	micro-vibration dampening
DO-007	guidance / bearing interfaces interfaces with fatigue-critical features register: signal energy amplification (dimensions, tolerances)	CONCEPTUAL	micro-vibration dampening
DO-008	Device boundaries: patient/fluid inlet and outlet termination: control/data points required by incorporation seal definition	CONCEPTUAL	dampening materials and structures
DO-009	Flow/force/signal path geometry realizing the mechanism is unknown	UNKNOWN	load dampening materials and structures
DO-010	Operating range for member slenderness ratio within which the mechanism holds its stated behavior	UNKNOWN	computed bounds for member slenderness ratio; verification
DO-011	Operating range for yield/ultimate strength within which the mechanism holds its stated behavior	UNKNOWN	computed bounds for yield/ultimate strength; verification
DO-012	Operating range for applied cyclic load amplitude within which the mechanism holds its stated behavior	UNKNOWN	computed bounds for applied cyclic load amplitude; verification
DO-013	Operating range for required run-out cycles within which the mechanism holds its stated behavior	UNKNOWN	computed bounds for required run-out cycles; verification
DO-014	Operating range for articulation radius within which the mechanism holds its stated behavior	UNKNOWN	computed bounds for articulation radius; verification
DO-015	Operating range for friction coefficient at guides within which the mechanism holds its stated behavior	UNKNOWN	computed bounds for friction coefficient at guides; verification
DO-016	No closed-loop control is proposed: the invention operates as an open-loop as specified; this record exists so the omission is explicit and auditable	CONCEPTUAL	
DO-017	Information path: sensing element -> decision/monitor or control/act (minimal); rich data paths are NOT APPLICABLE to this mechanical/physical invention	CONCEPTUAL	
DO-018	Fixture enabling: buckling / kink margin tests across duty cycle	UNKNOWN	rig concept named, build specific instrumentation selection; acceptance instrumentation
DO-019	Fixture enabling: fatigue life (run-out) testing — rig concept	UNKNOWN	build specific instrumentation selection; acceptance instrumentation
DO-020	Fixture enabling: friction and wear characterization — rig concept	UNKNOWN	build specific instrumentation selection; acceptance instrumentation

ID	Description	Status	Missing Inputs
DO-021	Fixture enabling: buckling/kink margin test across duty cycle	UNKNOWN	Unknown concept name; fixture specification; instrument selection; acceptance instrumentation

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
member slenderness ratio	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
yield/ultimate strength	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
applied cyclic load amplitude	UNKNOWN (no sourced value)	N	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
required run-out cycles	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
articulation radius	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
friction coefficient at guides	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
therapeutic or diagnostic failure	Failure by therapeutic or diagnostic error	design out of tolerance	USFDA 101 (requires design)	UNKNOWN, triggered and operating mode (Arstar)
kink/buckling below required	Failure by kink/buckling below required	compression buckling analysis	ASME 8 and 10, ASME 10.10.1.1	UNKNOWN, buckling load due to (Arstar)
fatigue crack initiation	Failure by fatigue crack initiation	stress concentration, fatigue crack growth	ASME 10.10.1.1, ASME 10.10.1.2	UNKNOWN, fatigue crack growth rate (Arstar)
wear debris generation	Failure by wear debris generation	material pairing and coating	ASME 10.10.1.1, ASME 10.10.1.2	UNKNOWN, wear debris generation rate (Arstar)
joint separation under force	Failure by joint separation under force	detent load and sizing	ASME 10.10.1.1, ASME 10.10.1.2	UNKNOWN, joint separation rate (Arstar)
set-release of superelastic	Failure by set-release of superelastic	shape memory alloy	ASME 10.10.1.1, ASME 10.10.1.2	UNKNOWN, set-release rate (Arstar)

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
therapeutic or diagnostic output failure	Failure of therapeutic or diagnostic output	design control and NOT ESTABLISHED (at this time)	design control and NOT ESTABLISHED (at this time)
kink/buckling below required duty	Failure by kink/buckling below required duty	pushability test and bend test and kink test	pushability test and bend test and kink test
fatigue crack initiation at stress concentration	Failure by fatigue crack initiation at stress concentration	stress concentration reduction	stress concentration reduction
wear debris generation at sliding interfaces	Failure by wear debris generation at sliding interfaces	material selection and lubrication	material selection and lubrication
joint separation under overload	Failure by joint separation under overload	load test and joint design	load test and joint design
set-release of superelastic shape	Failure by set-release of superelastic shape	set-release test and material selection	set-release test and material selection

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
VF-001	Subject pacemaker leads with and without pacing	Subject pacemaker leads with and without pacing	Subject pacemaker leads with and without pacing	Subject pacemaker leads with and without pacing
VF-002	run killer experiment: falsification_test	run killer experiment: falsification_test	run killer experiment: falsification_test	run killer experiment: falsification_test
VF-003	buckling / kink margin tests across duty	buckling / kink margin tests across duty	buckling / kink margin tests across duty	buckling / kink margin tests across duty
VF-004	fatigue life (run-out) testing. Acceptance	fatigue life (run-out) testing. Acceptance	fatigue life (run-out) testing. Acceptance	fatigue life (run-out) testing. Acceptance
VF-005	buckling/kink margin test across the duty	buckling/kink margin test across the duty	buckling/kink margin test across the duty	buckling/kink margin test across the duty
VF-006	fatigue run-out at worst-case loading	fatigue run-out at worst-case loading	fatigue run-out at worst-case loading	fatigue run-out at worst-case loading
VF-007	buckling / kink margin tests across duty	buckling / kink margin tests across duty	buckling / kink margin tests across duty	buckling / kink margin tests across duty
VF-008	overload margin demonstration to worst	overload margin demonstration to worst	overload margin demonstration to worst	overload margin demonstration to worst

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VA-001	end-use validation of verified design	NOT ESTABLISHED (defined after verification)	NOT ESTABLISHED (defined after verification)	NOT ESTABLISHED (defined after verification)

11. Materials

Component	Candidate Material	Source	Verification Required	Status
domain candidate material	nitinol / polymer composite membrane	minimally invasive device practice	True	ENGINEERING_PROPOSED

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
nitinol / polymer composite membrane	nitinol / polymer composite membrane	1	Material	nitinol / polymer composite membrane	NOT ESTABLISHED	NOT ESTABLISHED	Reliability verification required

13. Manufacturing

Status: NO MANUFACTURING ANALYSIS POSSIBLE — no design exists; processes are candidates only

Candidate Processes:

- laser-cut hypotube / braided composite shaft: domain registry manufacturing route - candidate only; no process qualification exists
- passivation and surface finish control: domain registry manufacturing route - candidate only; no process qualification exists

14. External Evidence

Source 1:

URL :

Excerpt:

15. Transfer Boundary

YOU RECEIVE:

- + Buyer receives: canonical invention specification (with epistemic classes).
- + Buyer receives: engineering specification (this document, all blocks classed).
- + Buyer receives: design input/output graph with explicit IDs.
- + Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).
- + Buyer receives: killer-experiment definition and its EIG basis.
- + Buyer receives: evidence custody references and hashes.
- + Buyer receives: unresolved-uncertainty register.
- + Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

YOU MUST DEVELOP:

- > Buyer must create: actual geometry and detailed design.
- > Buyer must create: sourced engineering parameters and acceptance thresholds.
- > Buyer must create: physical prototypes.
- > Buyer must create: all verification and validation results.
- > Buyer must create: manufacturing process qualification.
- > Buyer must create: regulatory pathway determination and submissions.

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-005', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'value of critical parameter member slenderness ratio [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-001'}
- {'unknown': 'value of critical parameter yield/ultimate strength [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-002'}
- {'unknown': 'value of critical parameter applied cyclic load amplitude [P_cr N]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-003'}
- {'unknown': 'value of critical parameter required run-out cycles [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-004'}
- {'unknown': 'value of critical parameter articulation radius [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-005'}
- {'unknown': 'value of critical parameter friction coefficient at guides [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-006'}
- {'unknown': 'acceptance criterion for verification VF-001', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-001'}
- {'unknown': 'acceptance criterion for verification VF-002', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-002'}

- {'unknown': 'acceptance criterion for verification VF-003', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-003'}
- {'unknown': 'acceptance criterion for verification VF-004', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-004'}
- {'unknown': 'acceptance criterion for verification VF-005', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-005'}
- {'unknown': 'acceptance criterion for verification VF-006', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-006'}
- {'unknown': 'acceptance criterion for verification VF-007', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-007'}
- {'unknown': 'acceptance criterion for verification VF-008', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-008'}

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	first physical article	maintaining the proposed subject	subject verification	NOT ESTABLISHED	NOT ESTABLISHED	NOT ESTABLISHED
WP-02	constraint-compliance envelope article	per method	constraint-compliance	NOT ESTABLISHED	constraint-compliance	NOT ESTABLISHED (to be quoted)
WP-03	article for VF-001	per method	Subject pacemaker lead	NOT ESTABLISHED	VF-001 verification	NOT ESTABLISHED (to be quoted)
WP-04	article for VF-002	per method	run killer experiment: lead	NOT ESTABLISHED	VF-002 verification	NOT ESTABLISHED (to be quoted)
WP-05	article for VF-003	per method	buckling / kink margin	NOT ESTABLISHED	VF-003 verification	NOT ESTABLISHED (to be quoted)
WP-06	article for VF-004	per method	fatigue life (run-out) test	NOT ESTABLISHED	VF-004 verification	NOT ESTABLISHED (to be quoted)
WP-07	article for VF-005	per method	buckling/kink margin	NOT ESTABLISHED	VF-005 verification	NOT ESTABLISHED (to be quoted)
WP-08	article for VF-006	per method	fatigue run-out at work	NOT ESTABLISHED	VF-006 verification	NOT ESTABLISHED (to be quoted)
WP-09	article for VF-007	per method	buckling / kink margin	NOT ESTABLISHED	VF-007 verification	NOT ESTABLISHED (to be quoted)
WP-10	article for VF-008	per method	overload margin demo	NOT ESTABLISHED	VF-008 verification	NOT ESTABLISHED (to be quoted)

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 3 governing equations and a 10-step build plan. The next experiment is clearly specified. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.